

Bendable Color ePaper Displays for Novel Wearable Applications and Mobile Visualization

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Figure 1: Our bendable color ePaper toolkit (A) enables novel, fully-functional wearable and mobile DataVis applications (B).

ABSTRACT

This paper presents a toolkit that allows to easily prototype with *bendable color ePaper displays* for designing and studying novel body-worn interfaces in mobile scenarios. We introduce a software and hardware platform that enables researchers for the first time to implement fully-functional wearable and UbiComp applications with interactive, curved color pixel displays. Further, we provide a set of visual and sensory-rich materials for customization and mounting options. To technically validate our approach and demonstrate its promising potential, we implemented eight real-world applications ranging from personal information and mobile data visualizations over active notifications to media controls. Finally, we report on first usage experiences and conclude with a research roadmap that outlines future applications and directions.

KEYWORDS

Color, ACeP, EPD, ePaper, E-Ink, electrophoretic, flexible, bendable, StrapDisplay, wearable, personal InfoVis, HCI, toolkit, mobileVis

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1 INTRODUCTION & BACKGROUND

Ultra-thin and flexible display technologies are key enabling factors for innovations in HCI. They make it possible to explore novel interfaces and visualizations that go beyond traditional form factors.

Therefore, manufacturing approaches for color-changing and light-emitting ambient displays using *chromic structures* (e.g., electro- [24, 46], thermo- [10], photochromic [25, 51]) or *luminescent materials* (e.g., electroluminescence [19, 26, 36]) have been investigated. While these are often characterized by single-color and segment-based or low-resolution designs, future wearable applications will also require pixel-based displays providing more dynamic capabilities. Unfortunately, flexible high-resolution OLED screens are rarely accessible, need constant energy and complex drivers (cf. [16]).

As a promising alternative, electrophoretic E-Ink/ePaper displays have been established as well-suited flexible, robust and bi-stable pixel screens. Prior research using bendable ePaper ranges from paper tabs [48], situated displays [18], foldable devices [17, 30, 37] over bracelets [9, 45, 49] and watch bands [29, 41] to bag straps [50] and on-clothing displays [11, 20, 28, 34]. While such examples show great potential, they, however, lack color since most ePaper displays *either* provide bendable capabilities *or* support color.

With this work, we present the first platform that allows to prototype with interactive, ultra-thin ePaper displays supporting both:



Figure 2: Our prototype consists of (1) a bendable color ePaper display, (2) three membrane potentiometers for touch input, (3) a curved and customizable housing and (4) a wireless multiprotocol, sensor-equipped microcontroller and an EPD driver.

bendable *and* color capabilities (see Figure 1, A). Therefore, we contribute an easy-to-use display toolkit that enables researchers to realize fully-functional wearable applications which are ready-to-use for real-world field studies. Further, we provide versatile customization options (e.g., surface materials, mounting) and implemented eight applications demonstrating the promising potential (see Figure 1, B1-B8). Finally, we conclude with a discussion for exploring future wearable and personal mobile data visualizations.

2 E-PAPER TOOLKIT & PROTOTYPE

In the following, we will provide a quick overview of all hardware components (see Figure 2) and introduce our software framework¹.

1 Display. Our prototype utilizes an organic active-matrix electrophoretic display (EPD) from Plastic Logic [31] with Advanced Color ePaper (ACeP) technology [13] providing four colors (red, blue, green, yellow) plus black and white. The 2.1" Legio display has a resolution of 240x146 pixels (132ppi), is bendable ($r=40\text{mm}$) and can be interfaced via SPI. A full-color refresh took 15 seconds.

2 Touch Input & Sensors. We decided for an explicit pressure touch requiring a small force for triggering actions. Inspired by StrapDisplays [29], we integrated three linear soft membrane potentiometers [47] that we placed behind the display. Besides, built-in sensors [1] extend the input with 9-DoF motion [42, 43] and ambient data (e.g., temperature [7], humidity [40], noise [44], light [8]).

3 Curved Housing. To securely enclose the display, we designed a curved case that can be easily 3D-printed and customized with sensory-rich materials such as leather, japan paper, or bindery cloth by (laser) cutting and gluing on custom cover templates. Versatile mounting options allow to wear the display with elastic bands, combine it with (smart) watches or use it as a stand-alone device.

4 Microcontroller. Due to its low energy and wireless Bluetooth capabilities, we build on the popular nRF52 SoC series [39]. Therefore, we combine the prototyping-friendly and feature-rich Adafruit Feather nRF52840 Sense [1] with an EPD driver [32]. To power the system up to 24 hours, we velcro-attached a 400mAh LiPo battery under the band which can be charged via USB (see Figure 1, A).

Connectivity. The prototype natively supports consumer media keys based on the HID profile. We further built in support for the Apple Notification Center Service (ANCS) [4] to bidirectionally

access many kinds of notifications (e.g., call, social, schedule, news, e-mail, health & fitness, location, entertainment) covering the most relevant real-world visualizations on smartwatch faces (cf. [23]).

App Development. To lower the barrier to entry, all apps are implemented within the Arduino ecosystem [2, 5]. To further simplify the development, we provide a suite of helper classes (e.g., AppManager, AppTemplate, Sensor- and ConnectionManager) as an overarching framework. Visual content is dynamically generated by using the Adafruit GFX [3] and PL_smallEPD [33] library, while Bluetooth services and sensor data can be subscribed on demand.

3 IMPLEMENTED EXAMPLE APPLICATIONS

To validate the technical feasibility and demonstrate the potential of our toolkit, we introduce eight example apps (see Figure 1, B1-B8):

B1 The *Activity App* shows a person's steps by using the inertial system [43] and visualizes a color-coded height profile (ascents in red) using the barometric pressure unit [7]. A blue arrow shows the steps in relation to the daily step goal on a colored progress bar.

B2 The *Weather App* displays the ambient temperature and humidity using built-in sensors [7, 40]. Further, we started to realize detail views with advanced visualizations (e.g., precipitation curves, conditions) and are working on an integration of real forecast data.

B3 The *DataVis App* explores visual encoding and interactive visualization techniques (e.g., starplots, scatterplots, charts, graphs) as generic components. We think that such colored widgets could positively impact research on ePaper data visualizations [29, 52].

B4 The *Watch+Strap App* seamlessly extends (smart) watches with additional, autonomous displays (cf. [29]). Since such applications typically require a strong coupling, we currently implement support for UI streaming [22, 29, 49] to enable distributed, hybrid interfaces.

B5 The *Notification App* demonstrates how incoming text messages from Apple mobile devices can be natively used for glanceable [6] and ambient usage scenarios using ANCS. Therefore, we realized an app that shows notifications with colorful and expressive emojis.

B6 The *Media Controls App* enables body-worn and aesthetic shortcuts for presentations, music player or remote shutter control. Due to its cross-platform implementation using HID profiles, it is possible to connect to any Bluetooth-enabled device without any drivers.

B7 The *Pattern App* presents an opportunity to turn the body-worn interface into a visual appealing and individually customizable accessory. In this regard, also user-defined inconspicuous interfaces

¹Detailed instructions and source codes are available online: <https://imld.de/epaper/>

(e.g., hidden progress bars, color-coded events) are possible (cf. [14]).

B8 The *Work Wear App* builds a basis for discussing new fields of application with domain experts. Therefore, we developed a set of representative work wear visualizations (e.g., hospital staff, construction worker, business analysts) for hands-on ideation sessions.

All apps serve us as a technological as well as conceptual foundation for further research. Additionally, they also could inspire new fields of application beyond wrist-worn devices and will help to quickly bootstrap new projects by building on the set of pre-programmed components (e.g., graphical widgets, sensor & wireless integration).

4 DISCUSSION & RESEARCH ROADMAP

While the presented toolkit still has some technical limitations (e.g., refresh rates, available colors, resolution, bend radius), we think that the constant improvement of ePaper technology will provide more advanced EPDs with larger color spaces, faster (partial) refresh options, and even more flexible properties within the next years. As next steps, we plan to move forward in several research directions.

» *Advanced Mobile Visualizations* show promising potential for wearables and smartphones (e.g., [12, 23, 35]). However, ePaper has been primarily used as a monochrome screen (extension) for novel interface assemblies using standard GUI components. Due to the lack of color, only few studies specifically focus on advanced ePaper visualizations. We think that the further investigation of aesthetic, meaningful and color ePaper data visualizations are yet underexplored, but important and promising building blocks for future wearable interfaces. Therefore, we plan to extend our DataVis app and study color-encoded, real-world data visualization in depth.

» *New Form Factors* are driving forces for novel interfaces. While we introduced a curved, wrist-worn wearable as a starting point, we are looking forward to go beyond the wrist and explore new, exciting form factors and flexible hybrid composites. As an example, we see promising potential in embedding EPDs with cutouts into garments to further blend on-clothing displays [11, 20, 24, 28, 34] and AR-attached visualizations [15, 27] into sensory-rich and traditional materials. Moreover, we envision a bright future for color-changing EPDs and films as a *display material* (cf. [46]) for accessories or interior designs using folded, non-planar and polygonal structures.

» *Conducting Field Studies* is a challenging, however, also very valuable method to gain empirical insights and investigate prototypes in their envisioned context of use [21, 38]. We aim to support research beyond the lab and developed a self-contained wearable toolkit. Therefore, we started to test our wrist-worn prototype on ourselves during a five-day hiking tour. We learned that the step counter works equally well as a smartphone, the EPD color slightly changes depending on the environmental temperature and a waterproofed housing is needed for rainy conditions. As a next step, we plan to implement logging and deploy research prototypes to end users to collect usage data and better understand possible design issues.

5 CONCLUSION

In summary, we hope to have contributed to the exciting research avenue of novel and innovative display technologies by presenting an accessible research toolkit for easy prototyping with interactive,

bendable and color ePaper screens. We described our feature-rich implementation of a curved, functional and colored wearable interface. Based on our design goals of an easy-to-use toolkit with self-contained, real-world functionality, the prototype is designed to enable development within the Arduino ecosystem, to natively support versatile wireless protocols and to be customizable by the user. To demonstrate the technical feasibility and value of the toolkit, we further introduced eight promising real-world applications as a foundation for further research. Finally, we took a glimpse into the future and discussed advanced visualizations, new form factors and field studies with forthcoming wearable applications making use of multi-color and flexible EPDs.

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